

GIS-based Emergency Fire Response System



*All One River GIS
Services LLC*

Background

- Sri Lanka is developing a nation-wide capability to use Emergency Response (911/119) systems to help in the management and response to emergency events. To accomplish this goal, private and public organizations will work together, including:
 - fire department and other emergency responders
 - local, provincial and national government
 - land-line and wireless telecom providers
 - consultants and engineering experts in 119, communications and mapping

International Partnership

Center for Conservation GIS (C2GIS)



**All One River GIS
Services LLC**

- Over 90 years of international experience
- Most recent emergency response GIS experience has been lead architect for the United States Department of Agriculture Emergency Management Response System (EMRS) for the Animal and Plant Health Inspection Service (APHIS).
- international GIS projects in South America (Argentina, Chile, Bolivia), Indonesia (Papua) and Northern Europe (Finland) for the energy, wireless telecom, and environmental consulting industries.
- over 18 years combined experience in field-related technical data collection and user-interface analysis and training.
- specializes in scanning, digitizing, hand entry, data verification, cataloguing, and data conversion.
- Clients include mining and public industry GIS users in Indonesia, Finland, Sweden, Mexico, and West Africa.

OBJECTIVE

- To develop a system
 - Where when a caller identifies the Colombo postal district (1-15) and cross street
 - It shows the closest fire station
 - the available water points, hospitals and other facilities
 - to the fire department operation center
 - this information is relayed to the closest station and the teams to take part in the emergency response.

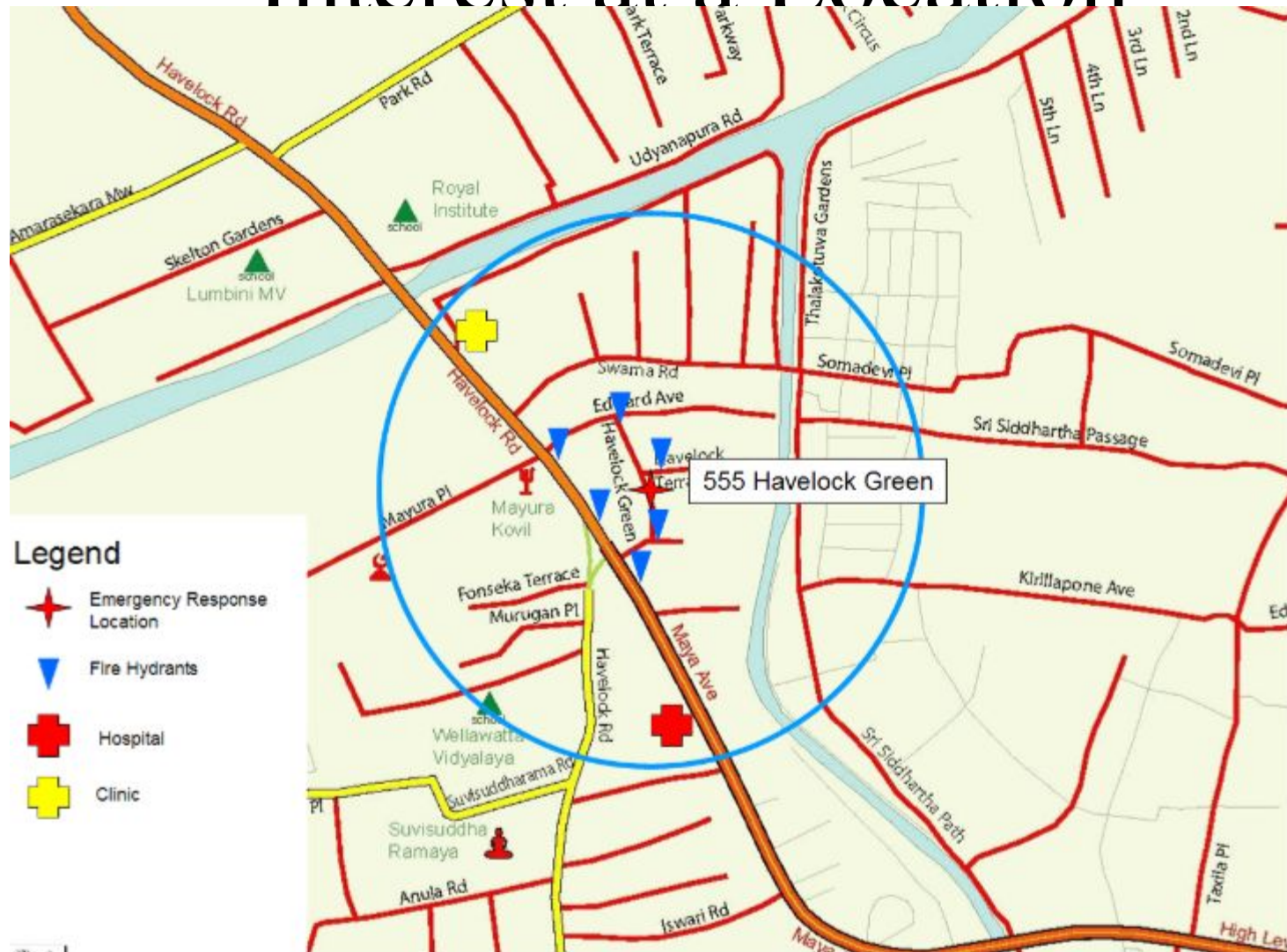
Examples

- *Doran –we need to include examples of screen shots from the similar systems you developed elsewhere. Video would be ideal.*



Street Map of Colombo

Interest at a Location



Task Scope

Task 1: Obtain all map street image files for Colombo.Approx. 50 sq. km.

Task 2: Digitize and populate attributes street names intersections for all street intersections in project area.

Task 3: Collect GPS points for all Emergency Fire Response System (EFRS) features by feature type: fire station, water tower, clinic, etc.

Task 4: Analyze geospatial points for closest EFRS feature (for each type) to each street intersection and build database relationship table.

Task 5: Analyze geospatial points for EFRS features in postal areas and build database relationship table.

Task 7: Deliver data and EFRS functionality compatible with ArcGIS format (.mxd).

Phase 2 - Provincial/National EFRS NEEDS ASSESSMENT

- The EFRS team (led by C2GIS) will review the Phase 1 system performance
- Conduct interviews with government regional and national representatives, representatives of the Sri Lanka Fire Department and Telecom officials
- Develop a plan for divisional/provincial and national implementation of the GIS-based Emergency Fire Response System.
- A summary report of the needs assessment

Phase 3 - ROLL OUT AT A Provincial/National LEVEL AND INTEGRATION INTO EMERGENCY PROTOCOLS.

- Approach, process and costs will be based on Phase 2 results.
- Will utilize and improve the systems used in Phase 1.

RESPONSIBILITIES

- It is ICET's / Fire Department's responsibility to
 - Organize permission from Ministry of Defense to use a GPS to collect the waypoints.
 - Arrange a vehicle/uniformed Fire officer with us to collect the points. We will need 10days for this.
 - A letter to be able to approach corporations which may have street level data for Colombo.
 - A review meeting every month
 - A team of 4 fire dept officials (with basic computer skills) to train in use of the system for 30 hours. They will also have approximately 60 hours of independent work prior to and during the implementation of the